

CHURN ANALYSIS AND VISUALISATION REPORT

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GitHub: *<https://github.com/nicnl31/customer-churn-tableau-python>*

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1. Introduction

Churn analysis is a mature field of research within the telecommunication industry, with established studies on understanding churning risk factors as well as detecting potential churners using modern machine learning techniques. [1], [2], [4] In telecommunication, churning is a popular issue for service providers, since offerings are typically on a prepaid subscription basis, [3] and customers have a wealth of choice from different service providers, so jumping between them to get the best deals is relatively easy. It is, therefore, in the responsibility of the Customer Relationship Management (CRM) team to deeply understand customers' current experience with the services offered and take steps to prevent cancellations before they happen. [4]

Preventing churn is important, mainly because of costs. It is shown that, in the telecommunication sector, the costs of acquiring a new customer can be up to six times greater than those of retaining an existing customer, [3] and the difference is even larger if there are more competitors within a market. [5]

This report presents a visual analysis of customer churn from a dataset originated from Maven Analytics. [6] The analysis is conducted to help the CRM team of our company address three pivotal problems in churn analysis:

Problem 1: Understand the company's performance in this quarter in terms of customer retention.

Problem 2: Understand the behaviours of churning customers.

Problem 3: Understand the characteristic differences between staying and churning customers.

The answer to these questions then guides the recommendations for business to tackle churn.

2. The data

The data contains information about 7043 customers in the second quarter of 2022. It features 38 fields, of which 22 are categorical and 16 are numeric. Summary statistics of columns of interest to the project are in Tables 1 and 2 in the Appendix. The dataset covers numerous angles including:

- Demographic information (e.g. age, gender)
- Location (e.g. city, zip code)

- Service information and statistics (e.g. contract type, subscriptions, charges, refunds, revenues)
- Marketing statistics (e.g. offers)
- Customer status: tenure in months, status (churned, stayed, joined).

This dataset is chosen because it offers a holistic view on the customer and can give answers to the stated problems based on both customer-side data (e.g. how much they spent, what services they are interested in) and telco-side data (e.g. what marketing offers we initiated). The time range of the data (i.e. within a single quarter) also corresponds to a typical churn analysis cycle usually performed in organisations, [7] so performing the analysis every quarter offers the company a regularly refreshed view of the customer.

3. Analysis Tools and Visualisation Methodology

The study uses two visualisation tools: Tableau and Python. Tableau is an industry-standard business intelligence tool, [8] with intuitive drag-and-drop operations enabling quick insight generation, and it is chosen for its simplicity of use and friendly interface. I also use Python [9] (in particular, its visualisation API such as Matplotlib and Seaborn), which is a general-purpose programming language, to perform more complex data processing and visualisation that may otherwise not easily be performed using Tableau.

To deliver meaningful visualisations and narratives, I will follow a two-step process with the help of these tools, while closely following the proposed research questions:

- *Exploratory analysis*: this is done in both Python and Tableau. I will use Tableau to create quick visualisations via dragging-and-dropping, to better understand the connection between variables. Python will be used to acquire summary statistics and description of the data, to quickly spot outliers/invalid data points and missing values. Since the research questions are about churners versus stayers, Python is also used to partition the dataset into two subsets for better analysis.
- *Explanatory analysis*: once better understanding of the dataset is acquired, visuals performed during exploratory analysis that carry the most insights are chosen for visual enhancements, using elements of pre-attentive processing, [10] such as colour and shape, and Gestalt grouping principles for perception processing, [11] such as continuity and closure. More targeted visuals, which go deeper into certain aspects of the analysis, are also created.

4. Findings

4.1. Customer Retention Performance in Q2 2022

Insight 1: [Figure 1] In Q2 of 2022, of all customers who are not known to be newly joined, 28.4% churned. This ratio represents almost one in four customers that joined the company.

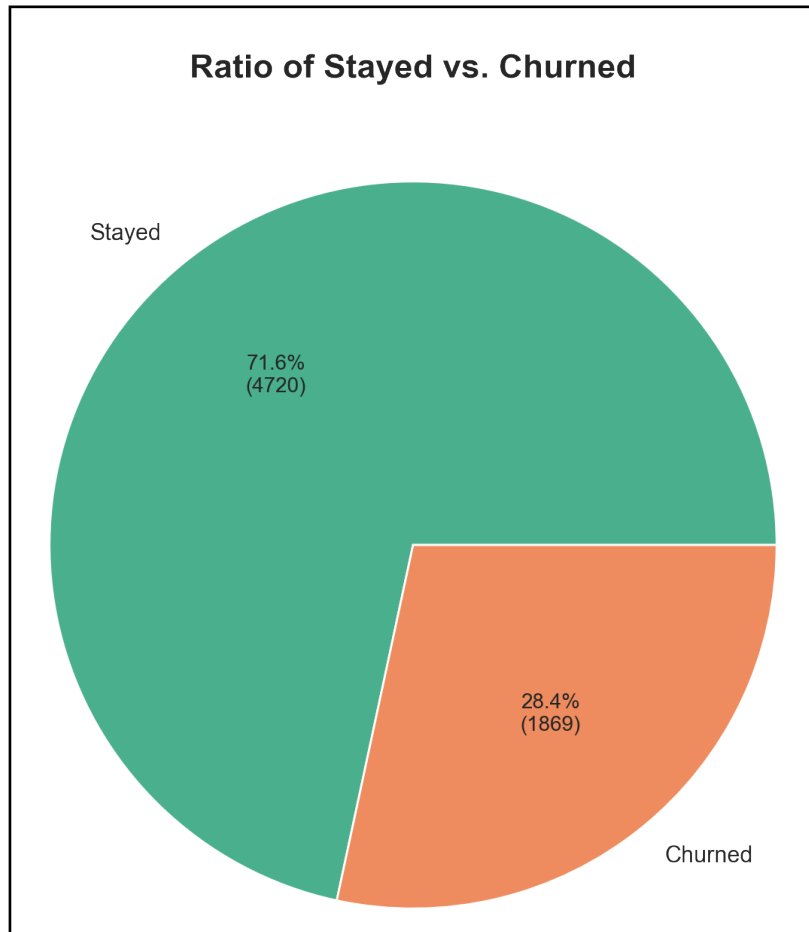


Figure 1: Ratio of stayed versus churned customers in Q2 of 2022. Created using Python.

4.2. The behaviours of churning customers

Insight 2: [Figure 2] Competitor is the number one reason customers are leaving the company. Of the five categories of churn, those who churned because of competitors account for 45% of all who churned.

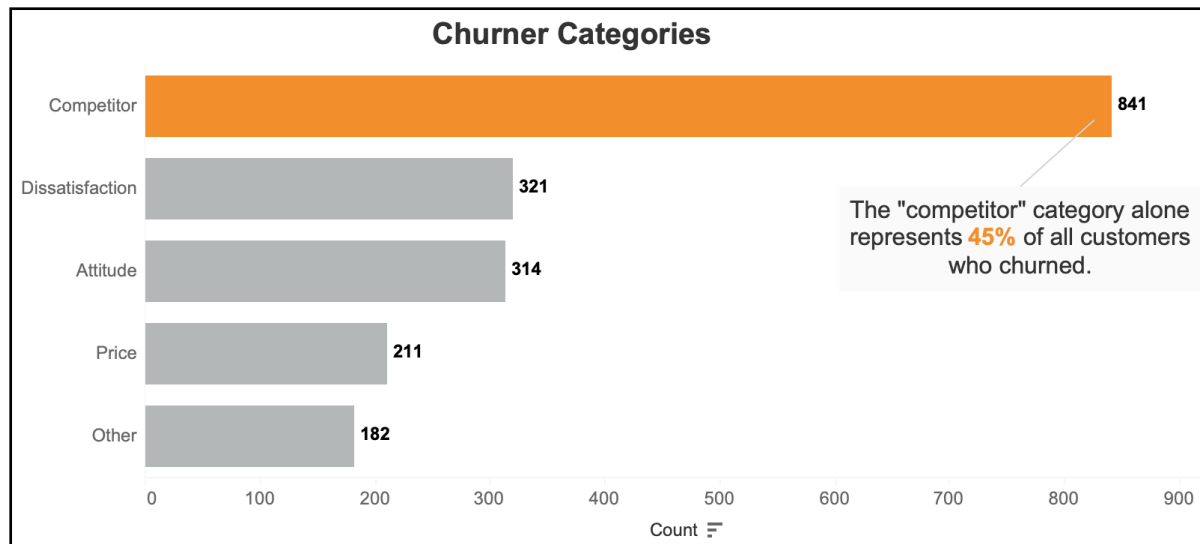


Figure 2: Churn categories bar plot. Created using Tableau.

Insight 3: [Figure 3] The top two reasons of customers leaving for competitors are to seek better offers and better devices from them.

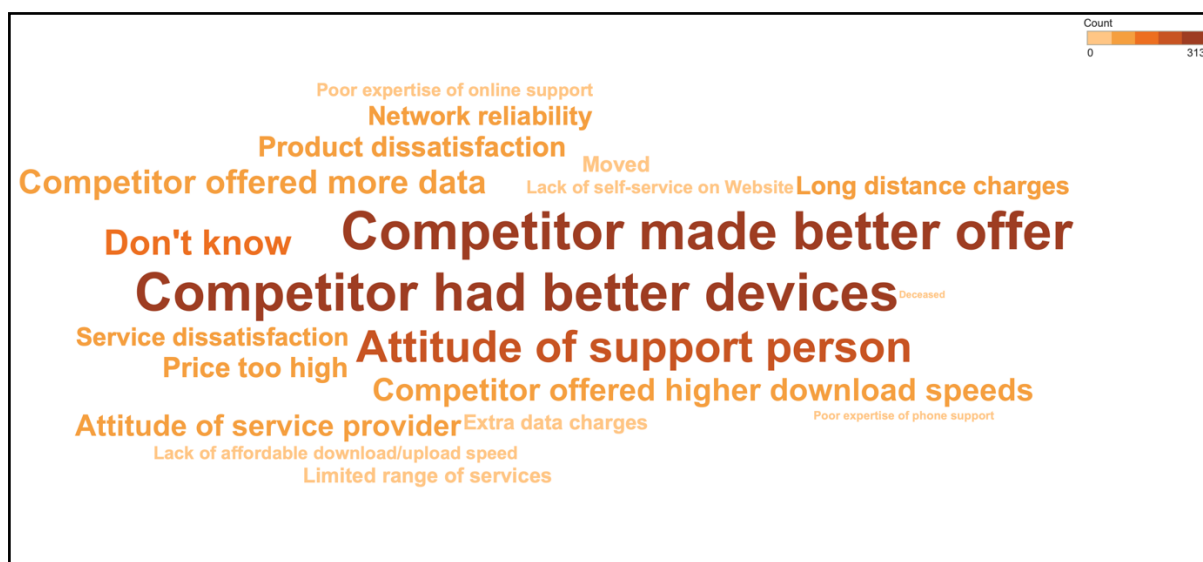


Figure 3: Churn reasons word cloud from customer feedback. Created using Tableau.

Insight 4: [Figure 4 + 5] Customers are churning considerably more at over \$65/month charge. There is also a cohort of customers taking low-priced offers (from \$18 – 21/month) and leaving after a short period of time (Figure 5 further highlights the mean and median tenure of this cohort). The mean tenure for this cohort is 6.85 months, while the median is just 1 month.

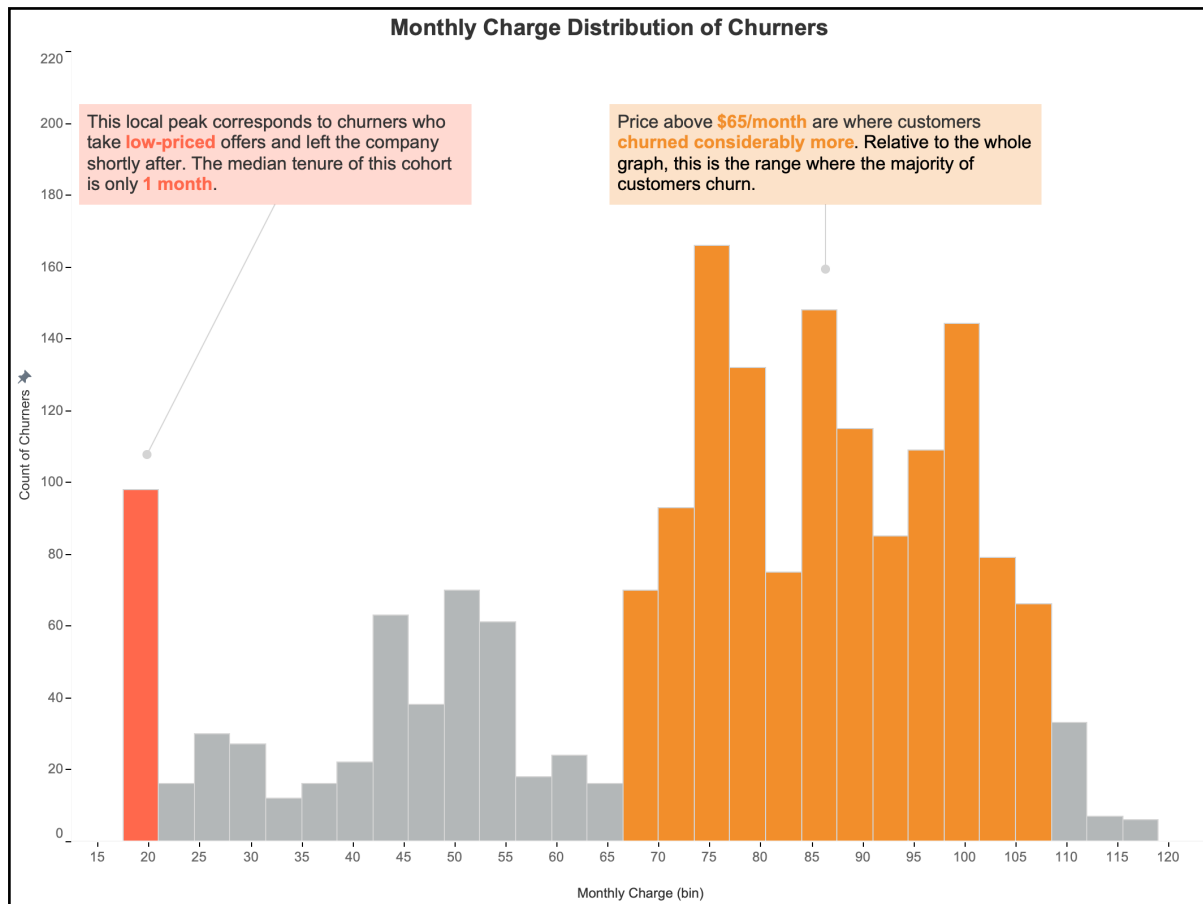


Figure 4: Monthly charge distribution of churners. Created using Tableau.

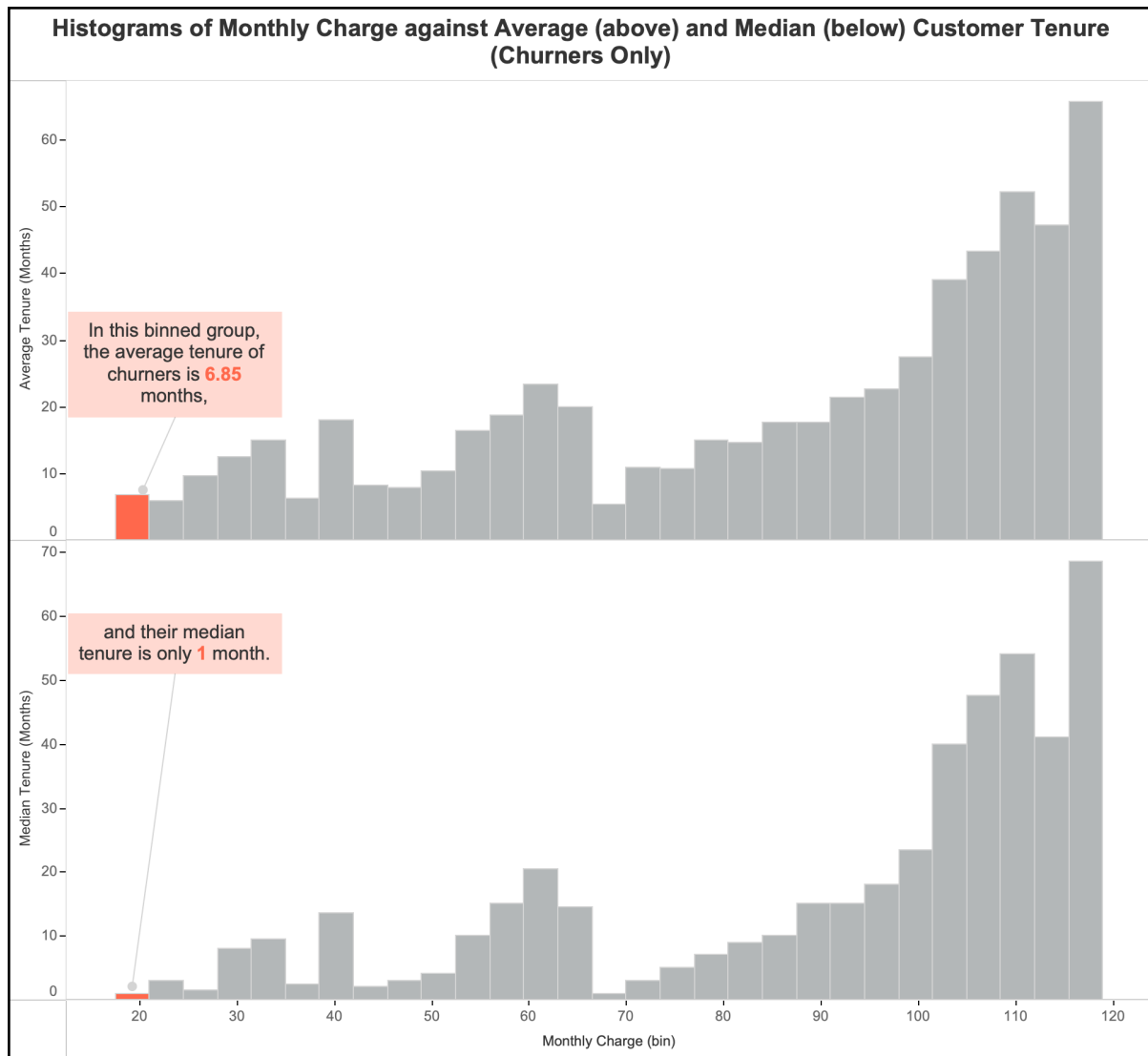


Figure 5: Monthly charge distribution of churners by average and median tenure. Created using Tableau.

Insight 5: [Figure 6] Most churners leave the company in the first five months of signing up.

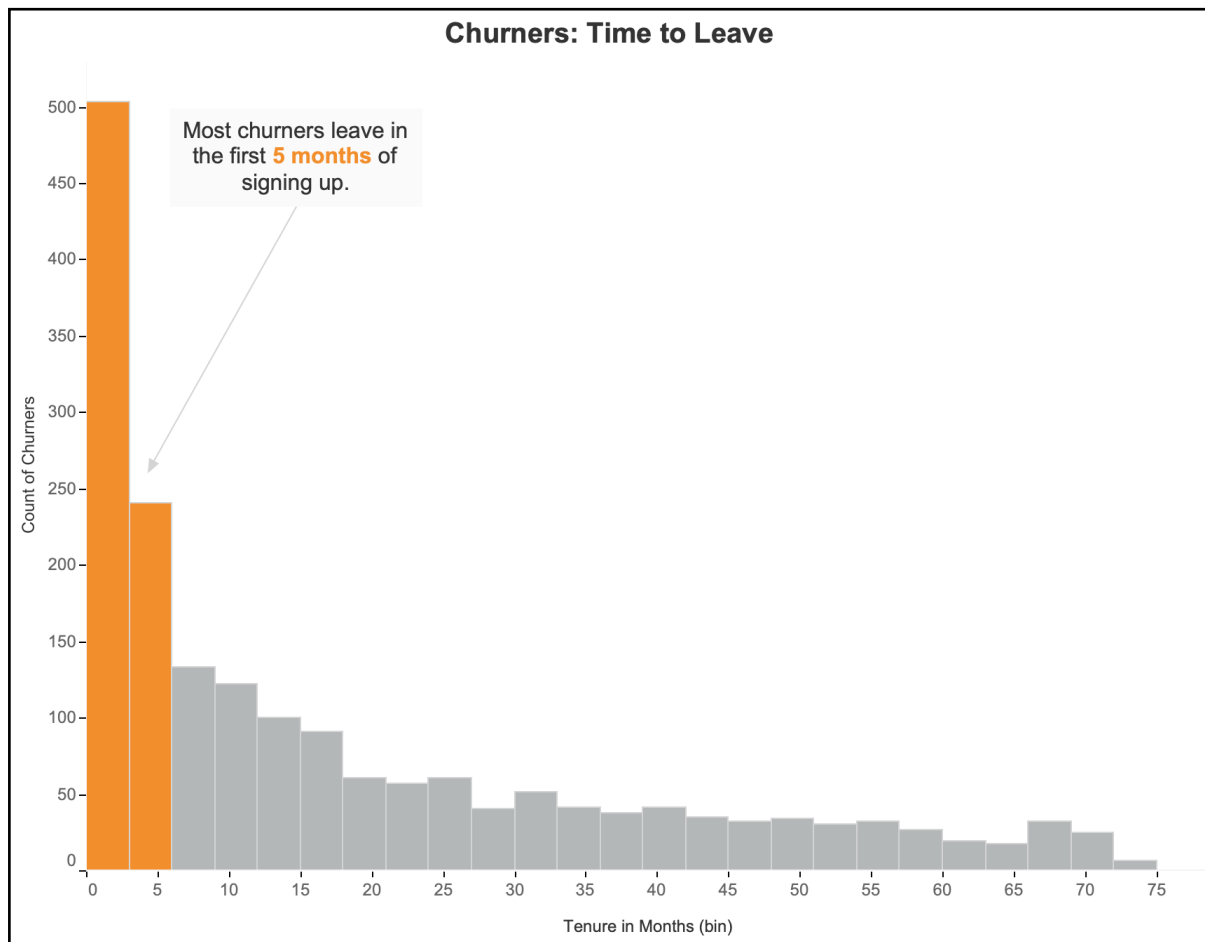


Figure 6: Time to leave for churners. Created using Tableau.

Insight 6: [Figure 7] In terms of the five marketing offers the company is running, Offer E leads to the greatest number of churners. It is also the only offer with more churners than stayers. Offer B is the best performing in terms of retaining customers.

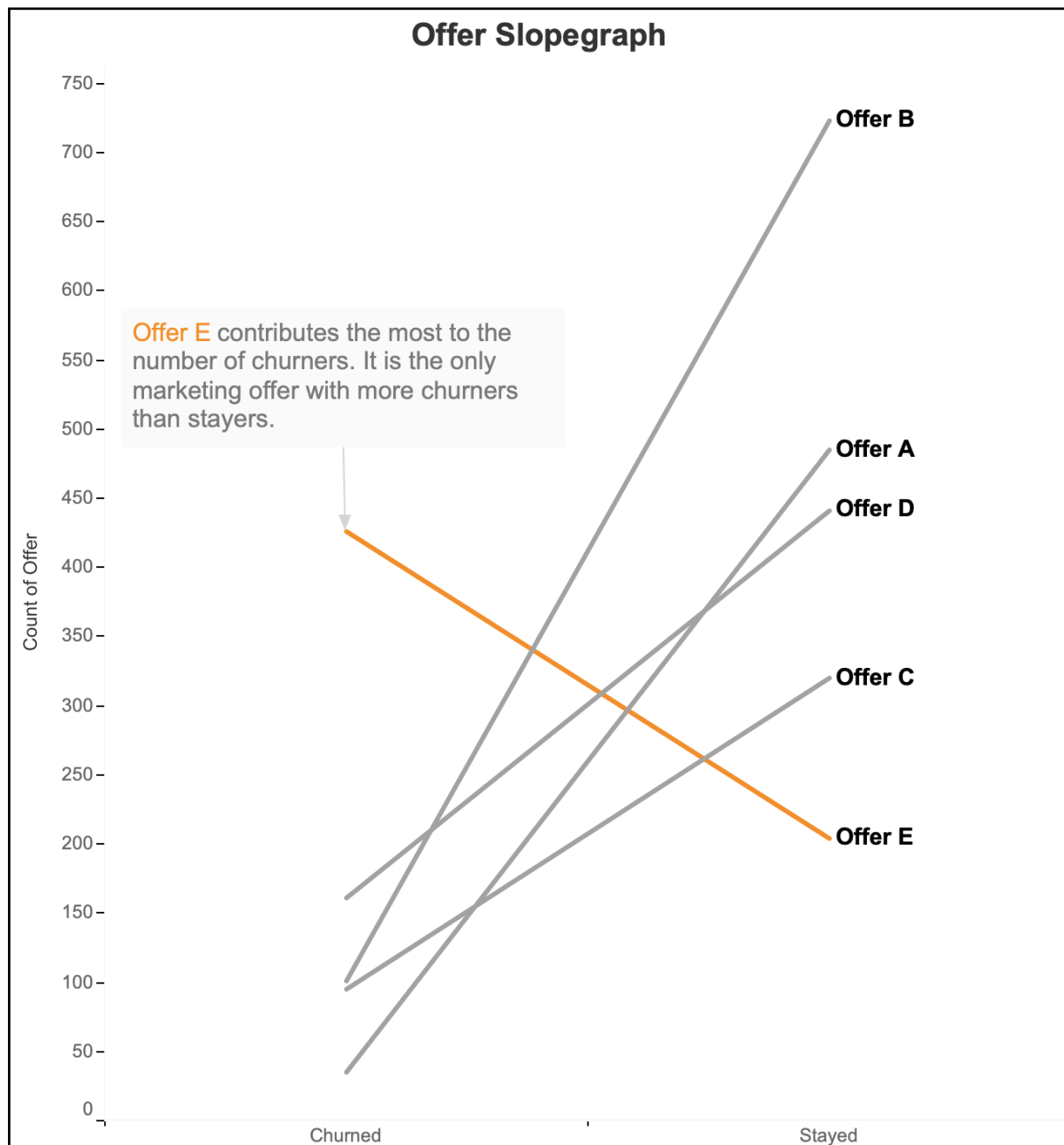


Figure 7: Slope graph of marketing offers. Created using Tableau.

4.3. Characteristic Differences between Stayers and Churners

Insight 7: [Figure 8] Customers aged 35 – 45 are most likely to stay, and customers aged 65+ are most likely to churn. Customers aged 18 – 25 are least likely to churn. This suggests a higher tolerance level in younger cohorts.

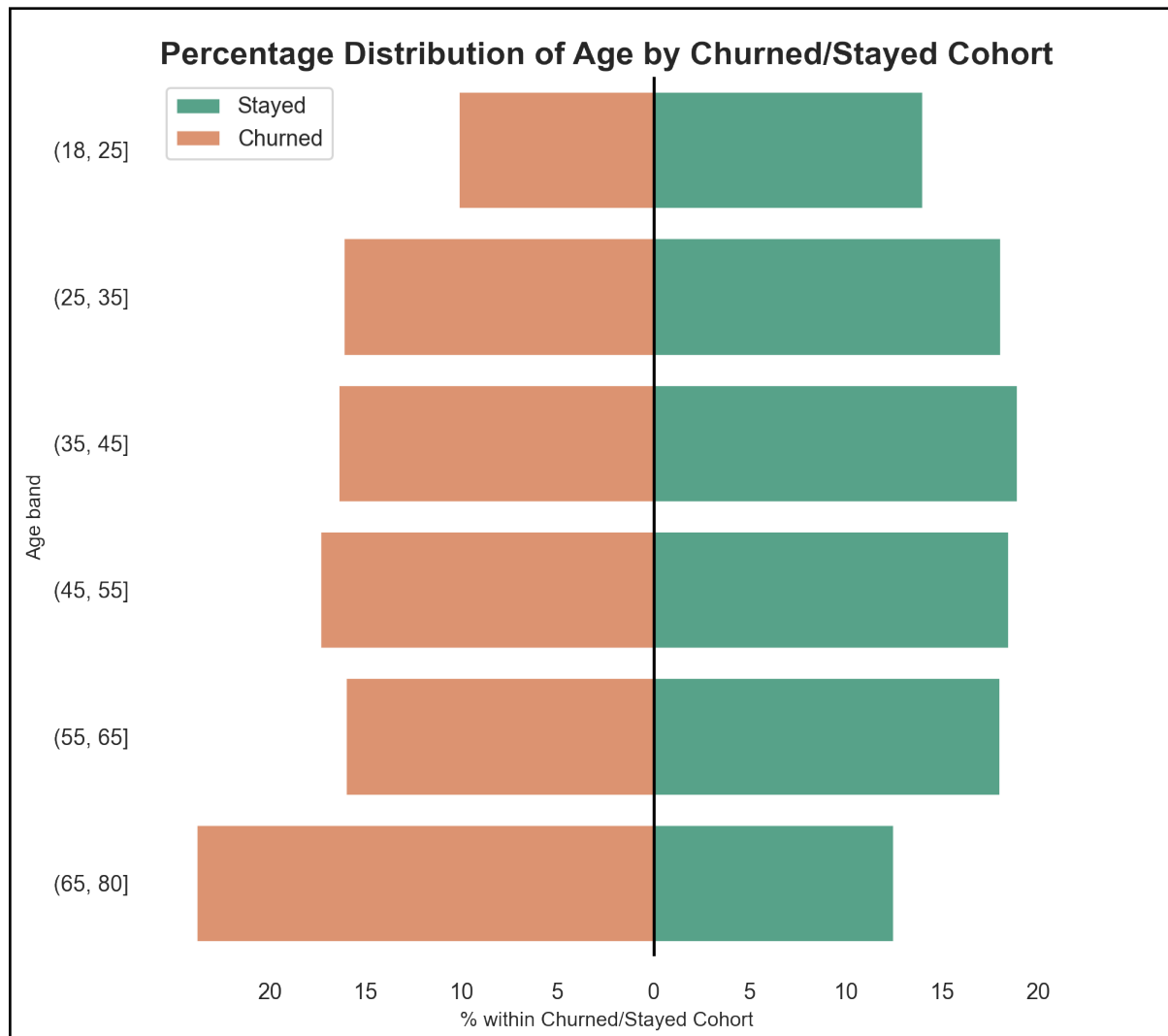


Figure 8: Age pyramid of churners and stayers. Created using Python.

Insight 8: [Figure 9] Male customers are slightly more likely to stay than female customers. There is not a significant difference in gender in those who churned.

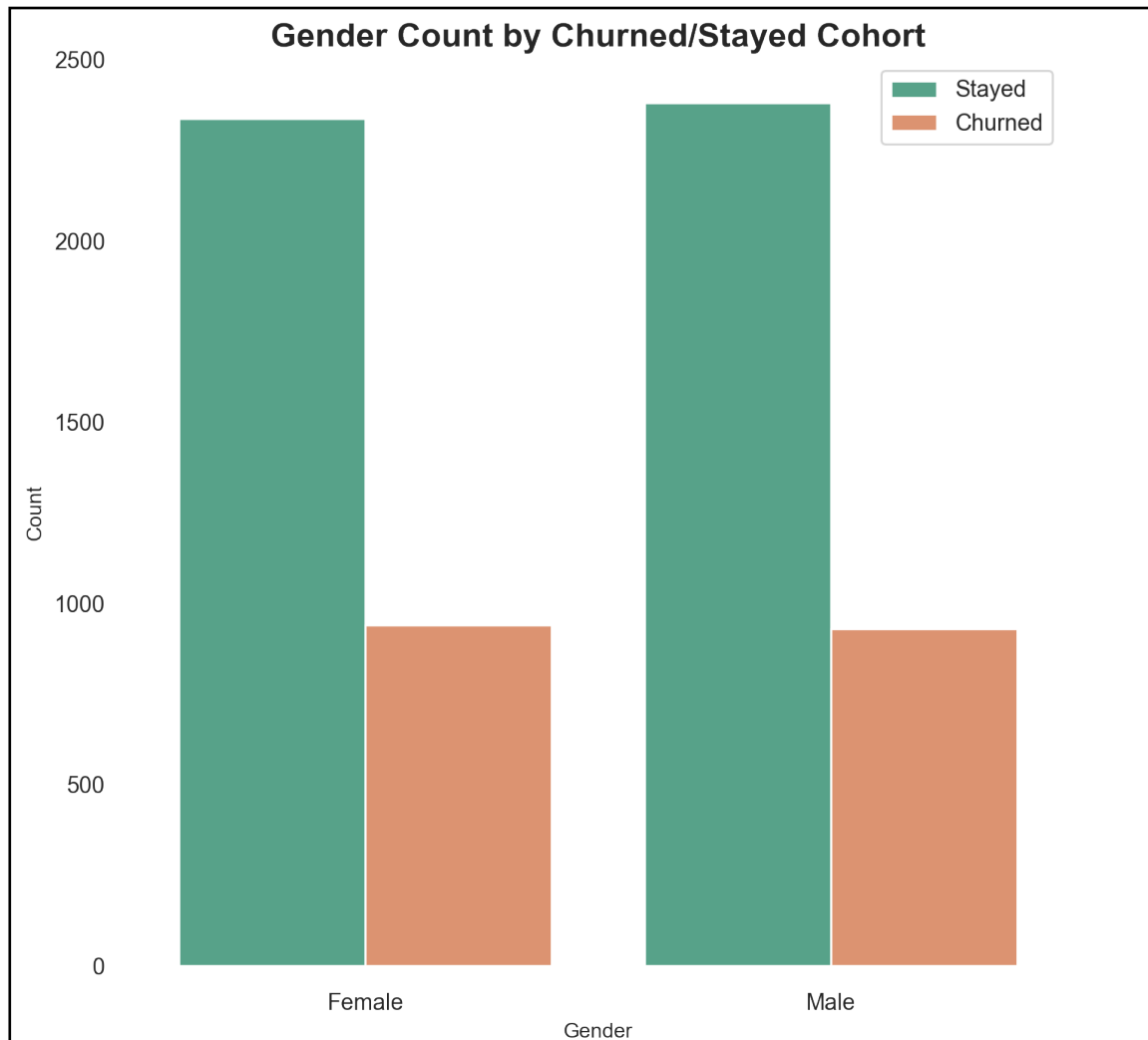


Figure 9: Gender count by churned/stayed cohort. Created using Python.

Insight 9: [Figure 10] In terms of charges, churners are paying more than stayers, both monthly (Subplot 1) and throughout their whole tenure. (Subplot 2) Churners are paying considerably more than stayers in extra data. (Subplot 3) Some long-staying churners (40+ months) are paying more long-distance charges than their stayed counterparts. (Subplot 4) Since charges are regressed on tenure, there is overall a positive trend in both stayers and churners for all subplots. (i.e. as they stay longer, they pay more)

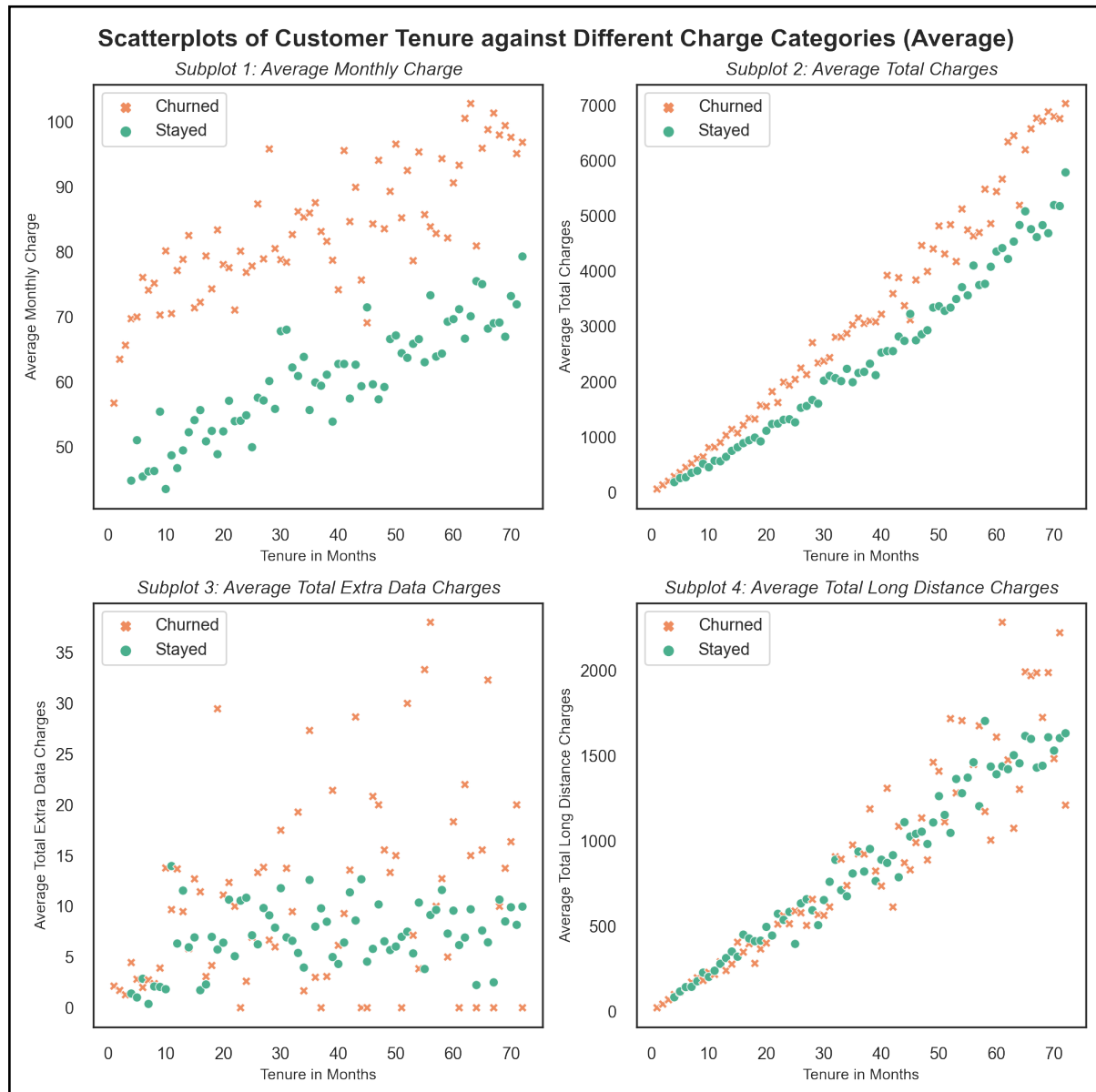


Figure 10: Customer tenure against different average charge categories. Each average is grouped by tenure in months. Created using Python.

Insight 10: [Figure 11] The majority of churners (first 2 churner bars in the revenue distribution) generated less than \$1,000 in total revenues, while the majority of stayers generated up to \$5,000.

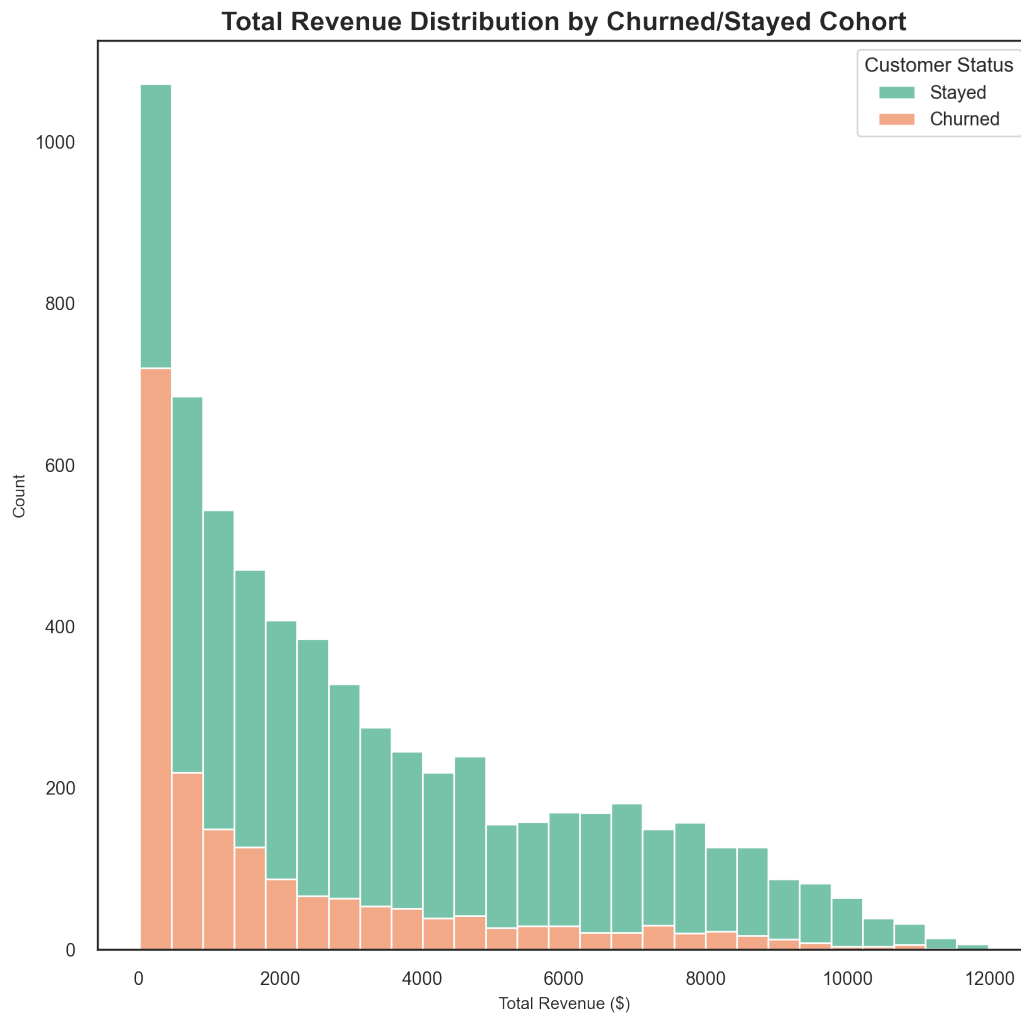


Figure 11: Revenue distribution by cohort. Created using Python.

Insight 11: [Figure 12] Stayers are referring more people to join than churners, on average. The slope of the trend line for stayers are steeper than that for churners, meaning that, on average, stayers also tend to refer more than churners as they stay longer.

Tenure in Months against Average Number of Referrals by Customer Status, plus Trend Lines with 95% Confidence Bands

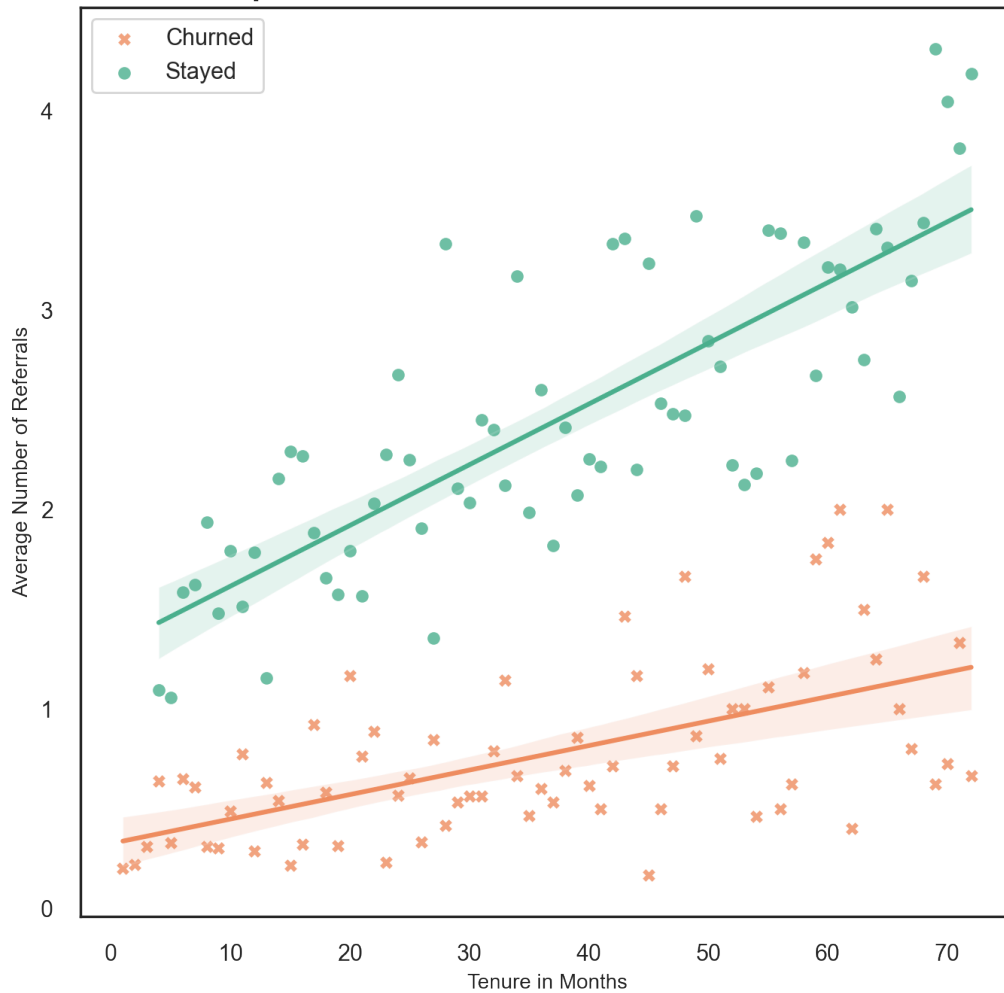


Figure 12: Tenure against average number of referrals. Created using Python.

5. Discussion and Recommendations

5.1. Addressing Problem 1

Compared to the 20 – 40% churn rate typical of the telecommunication industry, [12] our company is at the better end, (*Insight 1*) which signals good practices in churn management. However, it still has work to do to prevent customers from finding a better company, especially in terms of product/service pricing, (*Insights 2 – 5*) as it is evident most customers are leaving because prices are not attractive enough. A portion of customers may have been trying out our products and services, since they paid a small monthly amount (*Insight 4*) but left after a short while, most commonly after a month. This seems to agree with *Insight 5* that there is a peak in churners from very early on in their tenure, which suggests that customers may not have been very happy about the quality of products/services we offered, and there was no reason to stay.

5.2. Addressing Problems 2 – 3

The marketing department performed mostly well in retaining customers, (*Insight 6*) since most offers, except Offer E, attracted more customers than they pushed away. However, these offers did not seem to bring down excess charges, (*Insight 9*) which leads to churners paying considerably more than stayers. Also, the offers may not have covered enough monthly data, leading to some customers paying extra data charges.

In demographic terms, younger customers tend to be more forgiving, (*Insight 7*) while gender does not play an important role in predicting whether a customer stays or not. (*Insight 8*) Marketing can take these into account in their next campaign. Above all, improving products/services is always beneficial since it generates more revenue overall (*Insight 10*) and more organic traffic via referrals (*Insight 11*).

5.3. Recommendations

In the next quarter, it is recommended that the company reduce churn by:

- Reviewing quality of products/services that led to customers churning early on in their tenure.
- Implementing competitor analysis to understand current market pricings, in order to stay competitive.
- Directing Marketing to focus on younger cohorts for future campaigns, particularly those under 45.

- Offering more incentives for staying longer, such as extra data, distance charge waivers, and higher internet speeds. The same incentives can also be offered for referring customers to the service.
- Considering withdrawing Offer E from the market, and directing budgets to high-performing offers, such as Offer B.
- Reviewing staff training practices, and reviewing audio records of staff interactions with customers to address service quality concerns.

6. Conclusion

This analysis offered a thorough review of our company's retention performance, as well as the most pressing concerns for customers churning. It also reviewed characteristic differences between churners and stayers on aspects such as demographic information, services, and marketing, to offer a thorough view on customers and find possible improvements in the next quarter. These insights will greatly assist the company in reducing churn, as well as foster a more data-driven culture across departments.

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Appendix: Summary Statistics of the Dataset

Table 1: Summary statistics of numerical variables of interest to this project. Missing values have been filtered out. Invalid data points are highlighted in yellow, and their modified values are highlighted in green. Created using Python.

	Mean	Standard deviation	Minimum	Median	Maximum
Age	46.51	16.75	19.00	46.00	80.00
Number of Referrals	1.95	3.00	0.00	0.00	11.00
Tenure in Months	32.39	24.54	1.00	29.00	72.00
Avg Monthly Long Distance Charges	25.42	14.20	1.01	25.69	49.99
Monthly Charge	63.60	31.20	0.00 (-10.00)	70.05	118.75
Total Charges	2280.38	2266.22	18.80	1394.55	8684.80
Total Extra Data Charges	6.86	25.10	0.00	0.00	150.00
Total Long Distance Charges	749.10	846.66	0.00	401.44	3564.72
Total Revenue	3034.38	2865.20	21.36	2108.64	11979.34

Table 2: Summary statistics of categorical variables of interest to this project. Created using Python.

	Unique number of values	Most frequent value	Frequency of most frequent value
Gender	2	Male	3555
Offer	5	Offer B	824
Customer Status	3	Stayed	4720
Churn Category	5	Competitor	841
Churn Reason	20	Competitor had better devices	313